



A 6-MHz dual-phase valley-current AOT buck converter with virtual inductor current sensing and light-load efficiency optimization[☆]

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ABSTRACT

This paper presents a 6-MHz dual-phase valley current-mode adaptive on-time (AOT) buck converter for Li-ion battery-powered VRMs. To bypass active sensor bandwidth bottlenecks, a passive RC-based virtual inductor current sensing scheme is implemented, ensuring robust valley detection immune to PDN noise. A deterministic, PLL-less phase manager achieves precise 180° interleaving, eliminating low-frequency poles to preserve wideband stability. Additionally, an auto-zeroing zero-crossing detector (ZCD) facilitates seamless discontinuous conduction mode (DCM) transitions with sub-60-mA error, optimizing light-load operation. Fabricated in a 180-nm BCD process, the prototype utilizes miniaturized 120-nH inductors, achieving an 89.8% peak efficiency and a 17.7% efficiency boost at light loads. Under a 4-A load step, the converter recovers in 880 ns, yielding a 0.22-μs/A normalized recovery metric. Featuring high efficiency, extreme compactness, and fast dynamic responses, the proposed converter offers a reliable and ideal power solution for next-generation portable SoC applications.

1. Introduction

The escalating power requirements of artificial intelligence (AI) hardware and high-performance computing cores have necessitated voltage regulator modules (VRMs) with high power density, efficiency and transient agility. For portable systems operating from a 4 V input (e.g., Li-ion batteries), delivering a sub-1V supply involves high step-down ratios and large load currents, which impose stringent constraints on efficiency and dynamic voltage regulation [1–3].

To meet the stringent power density requirements of modern AI processors, pushing the switching frequency into the multi-MHz range (e.g., 5–10 MHz) has become a primary trend, enabling the use of miniaturized or even on-chip integrated passive filters—a concept often referred to as Power Supply on Chip (PwrSoC) [4]. However, operating at such extreme frequencies fundamentally challenges the control loop's integrity, as the shrinking switching period severely compresses the available time window for accurate signal acquisition and processing. High-frequency multiphase Buck converters are widely adopted to address these challenges, leveraging phase-interleaving to reduce output ripples and minimize passive components size. Among various control methodologies, valley current-mode adaptive on-time (AOT) control is particularly attractive. By permitting a momentary frequency increase during load steps, AOT control circumvents the bandwidth limitations

inherent in traditional fixed-frequency PWM schemes, achieving ultra-fast transient responses. Additionally, AOT architectures inherently support light-load efficiency optimization. The integration of a zero-current detector (ZCD) enables a seamless transition into discontinuous conduction mode (DCM) with autonomous frequency fold-back, effectively preventing reverse inductor current and mitigating conducting losses [5].

However, pushing the switching frequency beyond 5 MHz per phase significantly shrinks the VRM volume but introduces two critical control bottlenecks. First, achieving robust current sensing at such high frequencies becomes increasingly difficult. Traditional current sensing methodologies face severe trade-offs in high-frequency applications. For instance, SENSEFET-based sensing is highly susceptible to the significant switching noise generated during rapid power stage transitions. While DCR sensing is lossless, its implementation at high frequencies requires active amplifiers with exceptionally high gain-bandwidth products (GBW) to capture the inductor current ripple without significant phase lag [6,7]. This not only incurs substantial quiescent power consumption but also introduces non-negligible propagation delays, which provoke valley detection errors and degrade the stability of the inner current loop. Second, In multiphase systems, achieving precise interleaving without compromising loop bandwidth remains a critical challenge. While PLLs provide robust synchronization, they introduce

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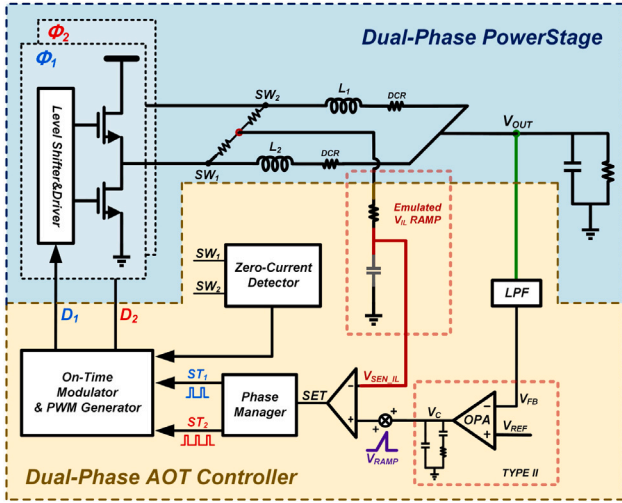


Fig. 1. System architecture of the proposed adaptive-on time (AOT) buck converter.

additional low-frequency poles and significant circuit complexity [8]. Recently, alternative approaches such as pulse distribution and analog phase interpolation have been proposed to mitigate interleaving jitters. For example, Tsai et al. utilized a master-slave phase interpolator to maintain stability at a 0.5 duty cycle [9]. However, these methods still rely on multi-phase analog ramp generators or comparators, resulting in additional area and power overhead.

To overcome these limitations, this paper presents a 6-MHz dual-phase AOT Buck converter featuring a simplified, high-robustness control architecture. Instead of relying on active amplifiers, the proposed design utilizes a RC network to emulate the inductor current ripple for valley current mode control. This purely passive virtual sensing scheme inherently bypasses active bandwidth bottlenecks, isolating the sensitive control loop from PDN interference and high-frequency switching noise.

For multiphase operation, this work adopts a logic-based Phase-Manager scheme instead of the PLL. By enforcing precise 180° interleaving through a deterministic digital distribution scheme, this architecture successfully avoids additional low-frequency poles, fundamentally preserving wideband small-signal stability. Additionally, an offset-cancellation based ZCD circuit enables seamless discontinuous conduction mode (DCM) transitions, effectively boosting light-load efficiency.

The prototype reliably operates across an input range of 3.3–5 V and an output range of 0.3–1.5 V. Experimental results validate a peak efficiency of 89.6%. Under a severe 4-A/200-ns load step, the converter achieves a tight 136-mV undershoot and a rapid 880-ns recovery time, verifying the extreme transient agility of the proposed AOT control scheme.

The remainder of this paper is organized as follows. Section 2 describes the proposed two-phase AOT architecture and its operating principles. Section 3 provides a detailed small-signal modeling and stability analysis of the proposed controller. Section 4 details the implementation of key circuit blocks, including the PhaseManager, AOT generator and the ZCD circuit. Section 5 presents the experimental results and performance benchmarks, followed by the conclusion in Section 6.

2. System architecture and principle

The overall architecture of the proposed two-phase AOT Buck converter is illustrated in Fig. 1. The system comprises a dual-phase power stage and a fully integrated controller optimized for high-frequency

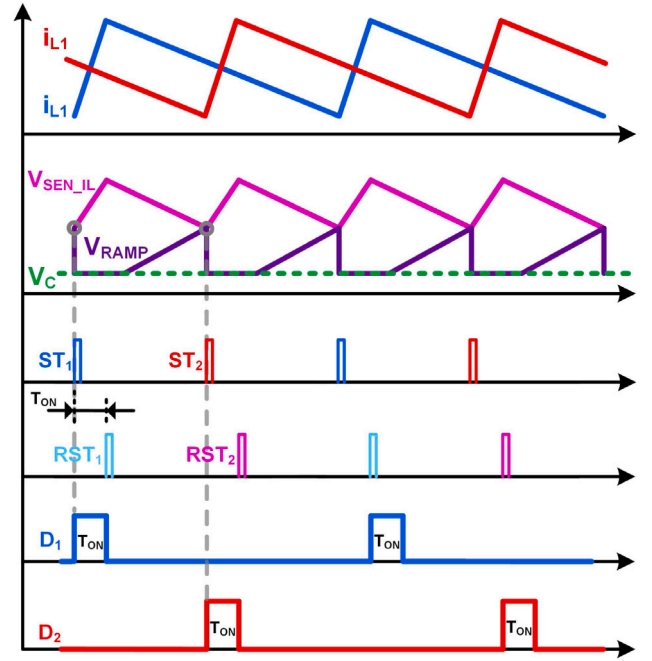


Fig. 2. Theoretical steady-state operating waveforms of the proposed dual-phase AOT controller.

operation (6MHz per phase). The voltage outer loop employs a Type-II compensator, which processes the error between the feedback voltage V_{FB} and the reference V_{REF} to generate the control voltage V_C .

2.1. Virtual inductor current sensing

Traditional DCR current sensing typically connects an RC network directly across SW and V_{OUT} [10]. However, in severe PDN parasitic environments, the high-frequency ringing noise from the power ground (PGND) can severely contaminate the sampled inductor current ripple, thereby interfering with precise comparator toggling. To overcome the robustness bottlenecks of traditional DCR sensing and eliminate active sensing delays at high switching frequencies, a simplified virtual current sensing scheme is implemented [11]. As illustrated in Fig. 3, it synthesizes an emulated ramp voltage synchronous with the aggregated inductor current. Rather than employing complex differential sensing across noisy power nodes, the proposed architecture extracts the current ripple directly from the switching nodes (SW_1 and SW_2) relative to the quiet analog ground (AGND). By routing both high-swing phase signals through matched resistors ($2R_S$) into a shared integrating capacitor (C_S), the passive network physically sums and low-pass filters the pulsed voltages. This ground-referenced RC integration yields the sensing voltage V_{SEN_IL} , which replicates the combined current information of both phases. This approach fundamentally shields the sensitive control loop from output-side PDN ringing and high-frequency switching residuals while significantly reducing the passive component count. To guarantee stability and suppress sub-harmonic oscillations, an external compensation ramp V_{RAMP} is superimposed on V_C .

Operating in valley current mode shown in Fig. 2, the main comparator monitors the intersection between V_{SEN_IL} and the ramped control signal $V_C + V_{RAMP}$. A master SET pulse is generated when the valley of V_{SEN_IL} drops below the peak of $V_C + V_{RAMP}$. Since the sensing signal aggregates current information from both phases, the master SET signal operates at twice the single-phase switching frequency ($2f_{SW}$). This aggregate (summed) sensing approach offers two critical advantages for high-frequency applications. First, it requires only one high-speed comparator to generate the master SET

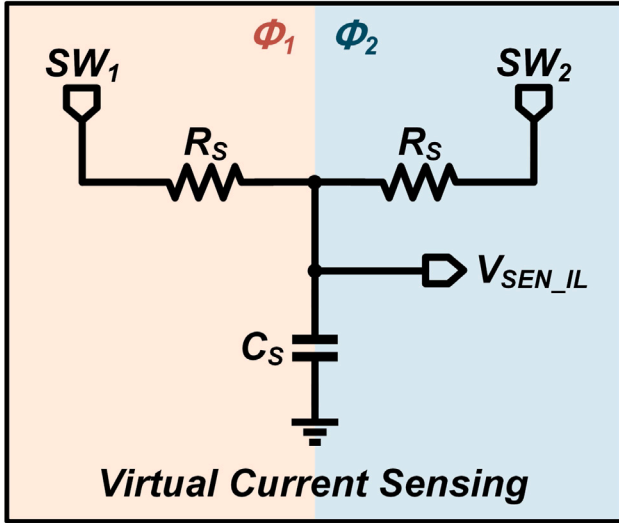


Fig. 3. Schematic of the RC filter-based virtual inductor current sensing network.

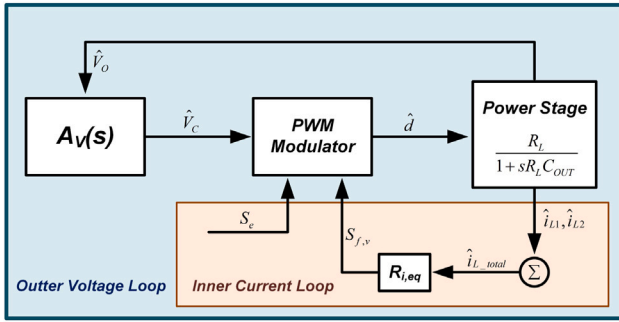


Fig. 4. Small-signal block diagram of the proposed dual-phase AOT converter with virtual current sensing.

pulse, drastically reducing the hardware and power overhead compared to single-phase sensing, which is conducive to low-power design. Second, the summed current allows the control loop to treat the two-phase power stage as a single equivalent inductor ($L/2$) [12]. This simplified plant dynamic enables the Type-II compensator to push the crossover frequency higher without complex inter-phase coupling management [13].

2.2. Phase manager and AOT generation

The generated master SET signal directly drives a logic-based Phase Manager. This module toggles the incoming pulses into two interleaved set signals (ST_1 and ST_2) with a rigorous 180° phase shift. Each set signal then triggers an independent AOT generator. To maintain a pseudo-fixed switching frequency across varying conditions, the AOT module adaptively scales the on-time (T_{ON}) proportional to the voltage conversion ratio (VCR). The resulting PWM signals drive the power stage via level shifters and buffers, ensuring balanced current distribution and minimized output ripple.

3. Small-signal modeling and stability analysis

The small-signal control architecture of the proposed dual-phase converter is illustrated in Fig. 4. The system utilizes a nested-loop control scheme, which is primarily composed of an outer voltage loop and an inner current loop featuring virtual current sensing. In the

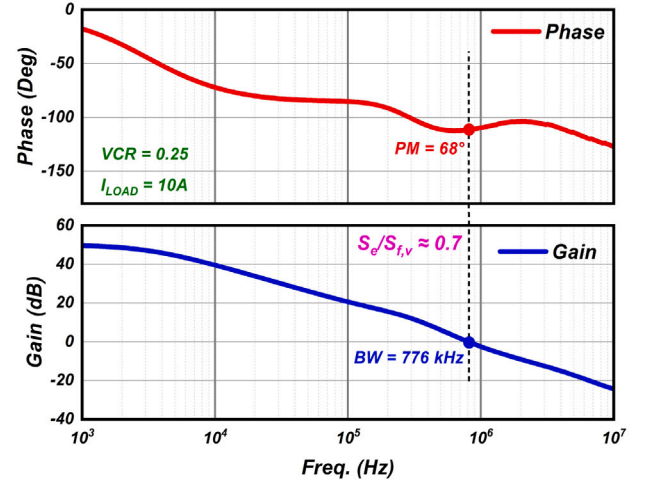


Fig. 5. Simulated closed-loop frequency response (Bode plot) under nominal operating conditions.

outer voltage loop, the output voltage perturbation $\hat{v}_o$ is processed by the compensator $A_V(s)$ to generate the control command $\hat{v}_c$. Simultaneously, the inner current loop aggregates the small-signal inductor currents from both phases ($\hat{i}_{L1}$ and $\hat{i}_{L2}$) to form the total current signal $\hat{i}_{L,total}$. This aggregate signal is then translated into the modulation ramp $S_{f,v}$ through the equivalent trans-impedance $R_{i,eq}$. Finally, the PWM modulator integrates the control voltage $\hat{v}_c$, the emulated current ramp $S_{f,v}$, and an external compensation ramp S_e to determine the duty cycle $\hat{d}$ for the power stage. Due to the extremely high switching frequency of 6 MHz, precise small-signal modeling is crucial for quantifying parasitic delay effects, suppressing sub-harmonic oscillations, and optimizing the loop bandwidth.

3.1. Modeling of RC-based virtual sensing

As established in Section 2.2, the ripple information of the inductor current is extracted through a RC-filter network. Unlike conventional DCR sensing, which directly connects an RC network in parallel across the inductor, the proposed virtual sensing architecture translates the high-swing signals from the power domain (SW and V_{OUT}) into the low-voltage control domain, thereby achieving efficient isolation from switching residual noise.

To accurately model the RC-based virtual current sensing network, its steady-state DC characteristics and AC small-signal assumptions must be explicitly defined. In steady-state operation, based on the fundamental inductor volt-second balance principle, the DC average of the switching node voltage (V_{SW}) equals the output voltage (V_{OUT}). Consequently, because the RC sensing network functions as a low-pass filter directly connected to V_{SW} , the DC component of the virtual sensing signal ($V_{SEN,IL}$) is naturally pinned to V_{OUT} .

Furthermore, to derive the high-frequency AC ripple component of the sensing network, it is assumed that the small-signal perturbation of the output voltage ($\hat{v}_{out}$) is negligible. This is based on the premise that the output capacitance (C_{OUT}) is sufficiently large to maintain a stable DC level, making the high-frequency voltage variation at the V_{OUT} node infinitesimally small compared to the large-signal voltage swing across the inductor. Under this assumption, the s -domain representation of the virtual sensing ripple can be derived as:

$$V_{SEN,IL}(s) = \frac{sL}{1 + sR_sC_s} \cdot [\hat{i}_{L1}(s) + \hat{i}_{L2}(s)] \quad (1)$$

At the target switching frequency of 6 MHz and within the required control bandwidth, the RC time constant is deliberately designed to

satisfy $sR_sC_s \gg 1$. In this implementation, the passive components are nominally set to $R_s = 50 \text{ k}\Omega$ and $C_s = 20 \text{ pF}$, yielding a time constant of $\tau = 1 \text{ }\mu\text{s}$. At the 6-MHz switching frequency ($\omega_{sw} \approx 37.7 \times 10^6 \text{ rad/s}$), the magnitude of $|sR_sC_s|$ is approximately 37.7, which strictly satisfies the high-frequency approximation margin. Under this high-frequency approximation, since the impedance of capacitor C_s is significantly lower than R_s , the RC integration network acts as an ideal current-to-voltage converter. Therefore, (1) can be approximated as a constant equivalent trans-impedance gain $R_{i,eq}$:

$$R_{i,eq} \approx \frac{L}{R_s C_s} \quad (2)$$

Following this derivation, the time-domain voltage slope $S_{f,v}$, which corresponds to the falling edge of the emulated sensing current, calculates strictly as $S_{f,v} = V_{OUT}/(R_s C_s)$. A paramount observation here is that the synthesized feedback slope relies exclusively on the regulated output voltage and the highly stable passive components (R_s and C_s). The formulation exhibits zero mathematical dependency on the actual physical inductance L . This fundamental decoupling mechanism drastically minimizes the vulnerability of the control loop to inductor magnetic saturation—a phenomenon where the effective inductance plummets during massive heavy-load transients [14]. By immunizing the feedback slope against inductance variations, the architecture secures consistent phase margin and transient fidelity, fulfilling the stringent stability requirements of the large-current VRM applications discussed previously.

3.2. Small-signal model of dual-phase modulator

In a valley current-mode architecture, SET pulse generation hinges on the intersection between the virtual sensing voltage and the ramp-compensated control voltage ($v_c + V_{RAMP}$). Because the logic-based Phase Manager deterministically distributes the aggregated SET signal across both phases, the entire modulator behaves dynamically as a single-phase converter switching at an equivalent frequency of $2f_{sw}$.

To accurately capture the sample-and-hold effect inherent in peak/valley current-mode control, the describing function method is applied [15]. By incorporating the virtual sensing network's transfer characteristics, we derive the continuous-time transfer function mapping the control voltage perturbation $\hat{v}_c$ to the total inductor current $\hat{i}_{L,total}$. Assuming negligible comparator propagation delays at 6 MHz, the aggregate modulator gain $F_m(s)$ is approximated as:

$$F_m(s) \approx \frac{1}{R_{i,eq}} \frac{1}{1 + \frac{s}{\omega_n Q_p} + \frac{s^2}{\omega_n^2}} \quad (3)$$

$$\omega_n = \frac{\pi}{T_{sw}/2} \quad Q_p = \frac{1}{\pi \left(\frac{S_e}{S_{f,v}} + 0.5 \right)} \quad (4)$$

Here, S_e represents the slope of the external compensation ramp, and $S_{f,v}$ denotes the falling slope of the virtual current ripple. Crucially, the interleaved parallel operation dictates that the equivalent switching period (T_{sw}) in the denominator halves the single-phase period.

The equations above expose a fundamental dynamic limitation: the current sampling mechanism inherently introduces a pair of high-frequency complex poles (ω_n) at half the equivalent switching frequency. These poles cause severe phase degradation and manifest physically as sub-harmonic oscillation susceptibility. Injecting an external compensation ramp S_e acts as a necessary damping factor (Q_p). Optimizing this $S_e/S_{f,v}$ ratio directly reduces Q_p , effectively suppressing the high-frequency resonance peak and damping the current loop. Conversely, an aggressively large S_e over-damps the system, deteriorating the control mechanism into pure voltage-mode operation and sacrificing the high-bandwidth transient advantage, thus dictates the fundamental trade-off between sub-harmonic immunity and transient agility.

Furthermore, to ensure loop stability against PVT variations, the fluctuation of on-chip passive components (typically around $\pm 20\%$)

must be evaluated. According to (4), a decrease in the $R_s C_s$ product (e.g., in the Fast-Fast process corner) increases the virtual ripple slope $S_{f,v}$. This variation reduces the effective compensation ratio $S_e/S_{f,v}$, which raises Q_p and the susceptibility to sub-harmonic oscillation. Therefore, a worst-case design strategy is adopted: the external ramp S_e is sized based on the minimum expected $R_s C_s$ boundary. Although this conservative sizing slightly over-damps the system at the typical corner, the complete elimination of active-sensor propagation delay in the proposed passive architecture allows the controller to still achieve a highly competitive crossover bandwidth.

3.3. Close-loop stability and compensator design

The overall open-loop gain, $T(s)$, cascades the Type-II compensator $A_v(s)$, the modulator gain $F_m(s)$, and the power stage plant. Leveraging the aggregated current sensing principle, the dual-phase power stage can be mathematically regarded as a single-phase equivalent plant with an effective inductance of $L_{eq} = L/2$. Accounting for the output capacitance C_{OUT} and its equivalent series resistance (ESR) R_L , the uncompensated loop gain is derived as:

$$T(s) = A_v(s) \cdot F_m(s) \cdot \frac{R_L}{1 + sR_L C_{OUT}} \quad (5)$$

Because the closed inner current loop functionally splits the LC double pole into two widely separated real poles, a simple Type-II compensator proves sufficient for robust voltage regulation. The compensator $A_v(s)$ places an integrator at the origin to drive steady-state DC regulation error to zero. A low-frequency zero is positioned to cancel the dominant load pole, while a high-frequency pole is placed strategically to roll off the gain before the switching frequency, suppressing high-frequency ripple feedthrough.

Fig. 5 plots the simulated Bode response under nominal load conditions. By calibrating the external ramp ratio $S_e/S_{f,v}$ to 0.7, the modulator provides adequate damping without starving the loop bandwidth. The properly tuned Type-II compensator ultimately delivers an aggressive 776-kHz crossover bandwidth alongside a robust 68° phase margin. This carefully architected frequency-domain profile guarantees absolute small-signal stability and aggressive transient tracking across the entire 0.3-to-1.5 V dynamic output voltage range.

3.4. Large-signal transient analysis

While the small-signal model ensures sub-harmonic stability, the large-signal load transient performance is fundamentally governed by the control architecture and the physical parameters of the power stage [16].

Compared to traditional fixed-frequency control schemes that exhibit inherent clock-edge latency, the proposed valley-current AOT control demonstrates a significant large-signal advantage. During a severe load step-up (ΔI_{load}), the output voltage droops abruptly. This droop propagates through the compensation network, causing the control voltage V_C to instantly rise above the virtual current sensing ramp. Because the proposed architecture operates in valley-current mode, the main comparator output remains continuously active, demanding the high-side switch to turn on.

Consequently, the AOT controller bypasses any clock delay and enters a transient saturation mode. It continuously triggers ON-time pulses (T_{ON}) separated only by the pre-defined minimum OFF-time ($t_{off,min}$) [17]. This $T_{ON} + t_{off,min}$ alternating sequence pushes the duty cycle to its maximum allowable physical limit, maximizing the inductor current slew rate.

Under this transient boundary condition, the effective inductor current slew rate (S_{eff}) is an average of the charging and discharging slopes over the saturated switching cycles:

$$S_{eff} = \frac{(V_{IN} - V_{OUT}) \cdot T_{ON} - V_{OUT} \cdot t_{off,min}}{L \cdot (T_{ON} + t_{off,min})} \quad (6)$$

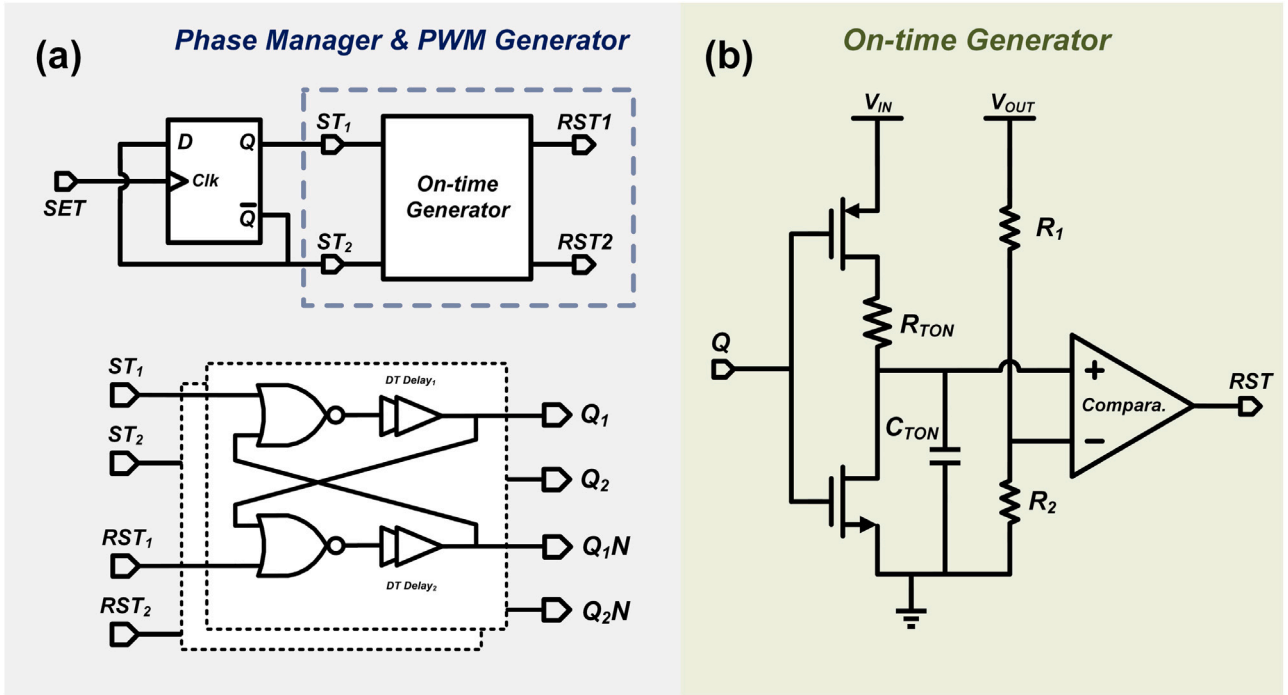


Fig. 6. Circuit implementation of (a) the PLL-less logic-based phase manager and PWM generator, and (b) the AOT generator.

Assuming the controller reacts instantaneously, the theoretical minimum recovery time (t_{rec}) and the corresponding maximum voltage droop (ΔV_{droop}) can be rigorously derived based on the charge balance principle:

$$t_{rec} \approx \frac{\Delta I_{load}}{S_{eff}} \quad (7)$$

$$\Delta V_{droop} \approx \frac{\Delta I_{load}^2}{2 \cdot C_{OUT} \cdot S_{eff}} + ESR \cdot \Delta I_{load} \quad (8)$$

where ESR is the equivalent series resistance of the output capacitor.

Eqs. (6) to (8) mathematically demonstrate that reducing the inductance L is the most effective approach to maximize S_{eff} , thereby minimizing both t_{rec} and ΔV_{droop} . By elevating the switching frequency to 6 MHz, the proposed design is optimized to utilize a small 120-nH inductor. Compared to conventional 1-MHz designs utilizing typical 1- μ H inductors, the 120-nH inductance accelerates the physical effective slew rate by nearly an order of magnitude. This physical power-stage optimization, perfectly synergized with the latency-free transient saturation mechanism of the AOT control, theoretically supports the extremely fast transient recovery performance observed in the experimental results.

4. Circuit implementation

To validate the proposed dual-phase valley current-mode AOT architecture, a 6-MHz dual-phase buck converter is designed and implemented. This section elaborates on the critical building blocks within the controller, specifically the phase manager, the AOT generator and the zero-crossing detector. The schematic implementations and design considerations are discussed in detail to elucidate the superiority of the adopted circuit techniques in achieving high-frequency stability and robust dynamic performance.

4.1. Phase manager and on-time generator

The integrated schematic of the Phase Manager, PWM Generator, and On-time Generator is illustrated in Fig. 6. As a common approach

in multiphase converters, precise phase interleaving relies on PLLs, which introduce significant power overhead and low-frequency poles that compromise high-frequency loop dynamics. To address this in the proposed 6-MHz architecture, a PLL-less, logic-based deterministic Phase Manager is implemented utilizing a high-speed toggle flip-flop (T-FF). Triggered directly by the master SET pulse operating at twice the switching frequency ($2f_{sw}$), the T-FF inherently performs frequency division to generate the phase-1 and phase-2 set signals (ST_1 and ST_2) with a rigorous 180° phase shift. This open-loop approach eliminates phase-error accumulation across PVT variations, minimizes the propagation delay to a sub-nanosecond scale, and introduces no additional poles or zeros, which significantly benefits the overall loop stability and reduces circuit complexity. The generated ST_1 and ST_2 pulses serve as the set triggers for the subsequent PWM Generator.

As shown in Fig. 6(a), the PWM Generator employs a cross-coupled NOR-gate SR latch, complemented by dead-time (DT) delay buffers, to safely drive the high-side and low-side power switches (Q and Q_N). Simultaneously, the onset of the conduction phase activates the corresponding On-time Generator (depicted in Fig. 6(b)) to determine the precise pulse width. The adaptive on-time is realized through an analog RC charging network. During the high-side conduction period, the input voltage V_{IN} charges the timing capacitor C_{TON} through the resistor R_{TON} . A high-speed comparator continuously monitors the capacitor voltage against the output voltage V_{OUT} . Once the charging voltage surpasses negative input of comparator, the comparator toggles to generate the reset signal (RST). This RST signal is immediately fed back to the NOR-gate latch to terminate the conduction phase and discharge C_{TON} . By dynamically coupling the charging slope to V_{IN} and the comparison threshold to V_{OUT} , this mechanism seamlessly achieves adaptive on-time control ($T_{ON} \propto V_{OUT}/V_{IN}$), effectively maintaining a pseudo-constant switching frequency across wide input and output variations without requiring complex phase-frequency detectors.

4.2. High accuracy zero-crossing detector

Entering DCM at light loads requires strictly blocking reverse inductor currents to prevent additional power loss of body diode conduction.

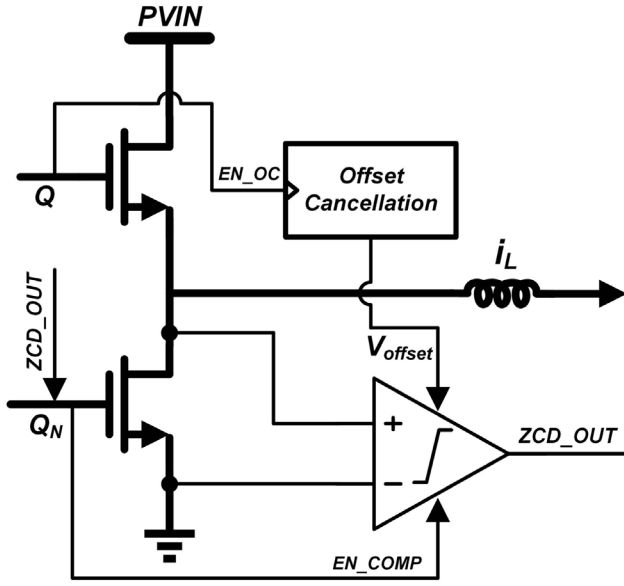


Fig. 7. Architecture of the proposed high-speed, auto-zeroing zero-crossing detector (ZCD).

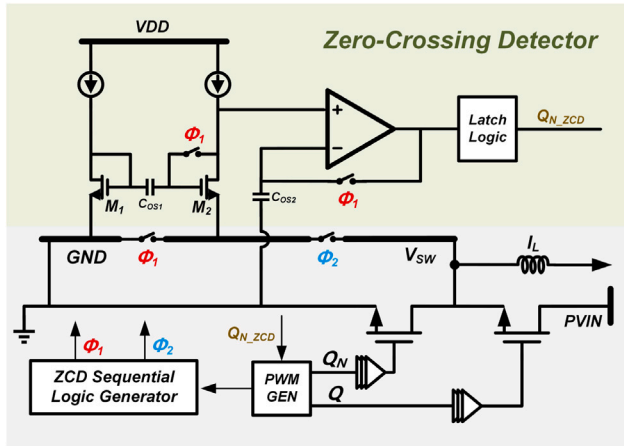


Fig. 8. Detail schematic of the two-stage-comparator-based auto-zeroing ZCD.

However, in a high switching frequency converter utilizing a miniaturized inductor, the current slew rate (di/dt) during the freewheeling phase is exceptionally steep. Consequently, even a nanosecond-scale propagation delay or a millivolt-level input-referred offset in conventional comparators provokes severe zero-crossing detection (ZCD) errors. To resolve this bandwidth-precision trade-off, Figs. 7 and 8 detail a high-speed, two-stage ZCD utilizing an auto-zeroing architecture to explicitly nullify static device mismatches and temperature-induced offset drift [5].

The proposed ZCD architecture leverages a common-gate pre-amplifier (M_1, M_2) with an active load. The circuit toggles between two strictly non-overlapping phases: offset memorization and zero-evaluation, governed by a custom sequential logic generator (Fig. 9). During the high-side conduction interval, the ZCD enters the memorization phase (Φ_1 closed, Φ_2 open). The input transistors M_1 and M_2 are diode-connected, forcing the second-stage operational amplifier (OA) into a unity-gain negative feedback loop. Under this configuration, the lumped input-referred offset voltage of the entire amplification chain – encompassing both the pre-amplifier mismatch and the OA static offset – is precisely extracted, sampled, and stored across the discrete auto-zeroing capacitors (C_{OS1}, C_{OS2}).

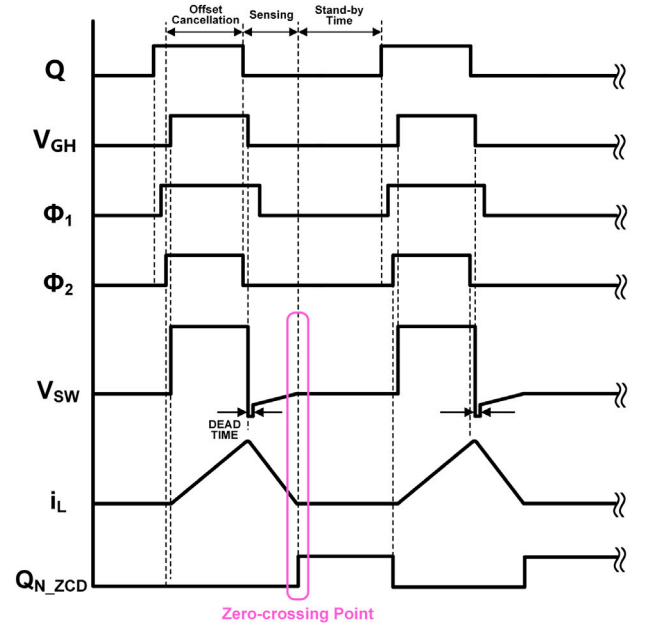


Fig. 9. Timing diagram of the offset cancellation and sensing phases of the auto-zeroing ZCD.

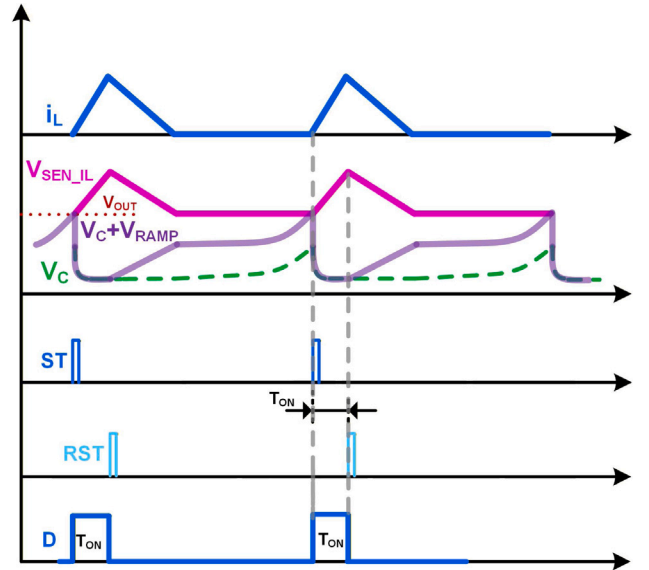


Fig. 10. Conceptual timing waveforms of the proposed AOT architecture in deep DCM.

Upon low-side switch turn-on, the ZCD immediately transitions to the evaluation phase (Φ_1 open, Φ_2 closed). The pre-amplifier now differentially senses the switching node voltage (V_{SW}) relative to the power ground. Because the memorized voltage dynamically shifts the DC bias to perfectly cancel the inherent static mismatch, the OA receives a pristine, zero-offset differential signal. While the forward inductor current (i_L) flows, the $I \times R_{DS(ON)}$ drop of the low-side switch pulls V_{SW} slightly below ground, maintaining the OA output at a logic high. As the decreasing inductor current approaches absolute zero, V_{SW} crosses the ground potential and swings positive. This exact zero-crossing violently trips the OA, forcing the comparator output (Q_N_ZCD) low. A dedicated high-speed digital latch instantly captures this state transition to shut off the low-side power FET, sharply truncating the reverse inductor current. Crucially, following the detection

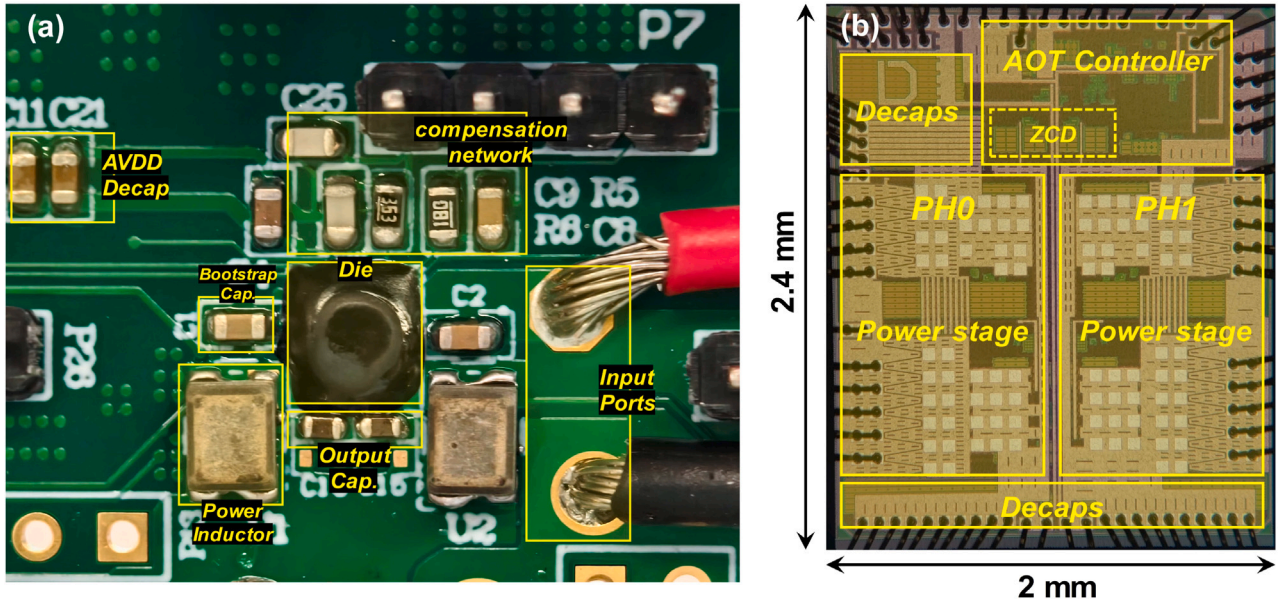


Fig. 11. (a) PCB evaluation setup, and (b) die micrograph of the fabricated dual-phase Buck converter prototype.

Table 1
Experimental parameters.

Parameters	Values
Input voltage range V_{IN}	2.5–5 V
Output voltage range V_{OUT}	0.3–1.2 V
Load range I_{load}	0–5 A/phase
Switching frequency f_{sw}	≈6 MHz/phase
Power inductor	120 nH/phase
Output capacitance	$2 \times 4.7 \mu\text{F}$

event, the ZCD asserts a micro-power stand-by mode. The tail current sources and biasing networks for both the pre-amplifier and the OA are aggressively cut off. This power-gating mechanism zeroes out the static quiescent current (I_Q) during the remainder of the switching period, thereby maximizing the overall converter efficiency under deep light-load conditions.

Beyond safely truncating the reverse current, the ZCD module intrinsically enables the autonomous frequency fold-back characteristic of the AOT architecture. Once the system enters DCM, the inductor current (i_L) rests at zero. Consequently, the AC component of the emulated virtual sensing voltage (V_{SEN_IL}) discharges to zero, leaving the signal clamped at the DC output voltage (V_{OUT}). Concurrently, the ZCD action cuts off the charging current for the compensation ramp, freezing V_{RAMP} at a constant DC level that is superimposed on the error control voltage (V_C).

To intuitively illustrate this mechanism, the conceptual timing waveforms under DCM operation are plotted in Fig. 10. During the extended freewheeling phase, the controller simply idles. As the load continuously discharges the output capacitor, V_{OUT} droops, causing V_C (and thus $V_C + V_{RAMP}$) to gradually rise. The main loop comparator remains active and continuously monitors these signals. The system naturally initiates the next ON-time the exact moment the rising $V_C + V_{RAMP}$ intersects the flattened V_{SEN_IL} , triggering a new *SET* pulse.

This mechanism explicitly demonstrates that the fundamental PWM triggering condition ($V_C + V_{RAMP} > V_{SEN_IL}$) remains structurally and mathematically identical in both CCM and DCM. Therefore, the architecture natively achieves seamless cycle-by-cycle frequency fold-back without the need for additional mode-switching logic or artificial hysteresis windows.

Ultimately, this autonomous frequency reduction drastically minimizes switching and gate-drive losses, yielding the pronounced light-load efficiency boost observed in the measurements.

5. Measurement results

In this work, a dual-phase buck converter prototype was fabricated utilizing a standard 180-nm BCD process. The printed circuit board (PCB) evaluation setup and the corresponding chip micrograph are presented in Fig. 11(a) and (b), respectively. To mitigate phase current mismatch and optimize power routing, the power stage features a strictly symmetric layout, which is essential for uniform thermal distribution and balanced dynamic performance. The fully integrated chip, including wire-bonding pads, occupies a total die area of $2.4 \times 2 \text{ mm}^2$. Key experimental parameters and component values are summarized in Table 1. Unless otherwise specified, all subsequent measurements were conducted under a nominal standard conversion condition from a 4-V input to a 1-V output, representing a typical step-down ratio required in Li-ion battery VRM applications.

Fig. 12 illustrates the steady-state switching node voltages and inductor current waveforms of the converter. During heavy-load continuous conduction mode (CCM), the system demonstrates a robust pseudo-fixed switching frequency of approximately 6 MHz, as depicted in Fig. 12(a). The dedicated phase manager successfully maintains a precise 180° interleaving between the two phases, effectively achieving ripple cancellation and thereby easing the requirements on the output filter capacitors. To extend battery life during standby or low-power states, the controller integrates a high-precision zero-current detection (ZCD) module. As shown in the discontinuous conduction mode (DCM) waveforms in Fig. 12(b), when the load current decreases to 200 mA, the AOT controller automatically scales down the switching frequency to 580 kHz. The high-speed ZCD comparator accurately identifies the inductor current zero-crossing point with a stringent detection error of less than 60 mA. This precise turn-off of the low-side switch effectively eliminates reverse current flow from the output capacitor to the ground, significantly reducing conduction losses and preventing unwanted sub-harmonic ringing at the switching node.

The dynamic performance of the proposed converter was comprehensively evaluated under rapid load-step transients (200 ns transient time), as captured in Fig. 13. During a severe step-up transient, the output voltage exhibits a minimal undershoot of only 136 mV, while a corresponding load step-down results in an overshoot of 275 mV. Notably, the converter achieves an ultra-fast recovery time of just 880 ns following the load step-up. This transient measurement clearly highlights the inherent architectural advantages of the proposed AOT

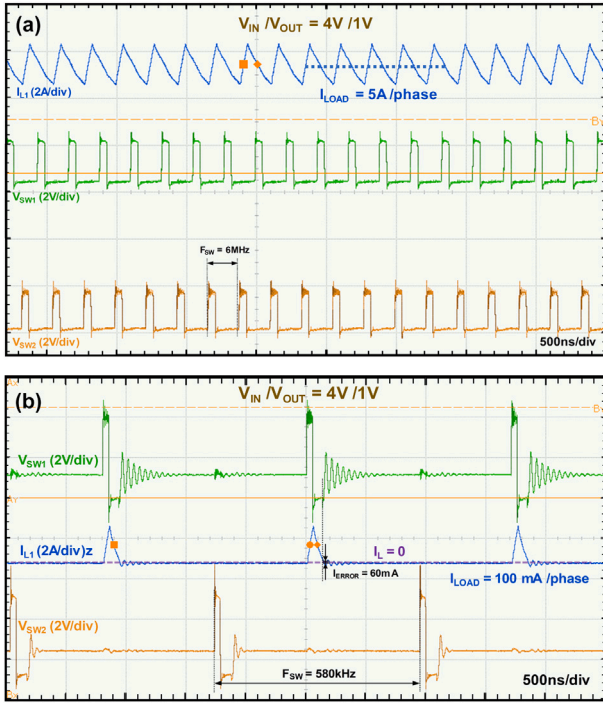


Fig. 12. Measured steady-state waveforms of the switching nodes and inductor current in (a) continuous conduction mode (CCM) at 6 MHz, and (b) discontinuous conduction mode (DCM) at 580 kHz with the ZCD enabled.

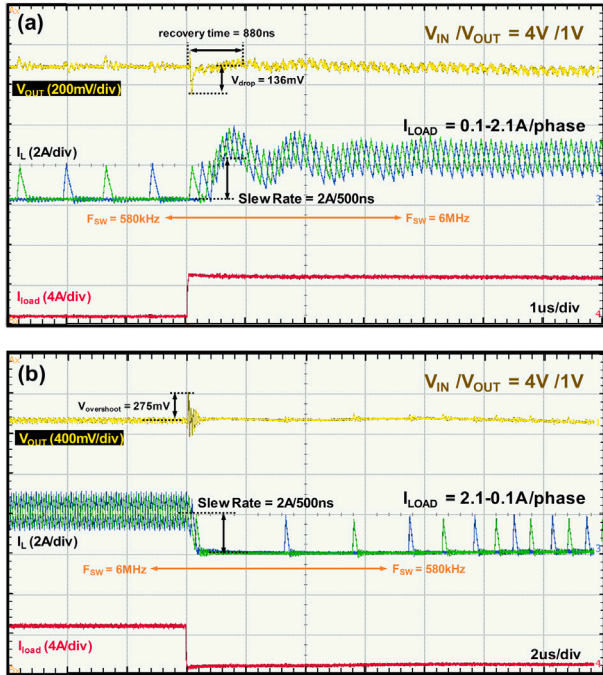


Fig. 13. Measured load transient response under load step highlighting seamless mode transitions: (a) step-up from 0.2 A to 4.2 A (DCM to CCM), and (b) step-down from 4.2 A to 0.2 A (CCM to DCM).

control scheme. Unlike traditional fixed-frequency control methods that are strictly constrained by loop bandwidth, the AOT controller permits a momentary, instantaneous increase of switching frequency during load transient, drastically accelerating the transient recovery process. Furthermore, with the ZCD module enabled, the system seamlessly

Table 2

Summary performance.

Technology	0.18- μ m BCD
No. of phases	2
Die size	$2 \times 2.4 \text{ mm}^2$
Peak efficiency	89.6% @ $V_{IN}/V_{OUT} = 4V/1V$, $I_{load} = 1.5A/phase$
Light load efficiency	85.3% @ $V_{IN}/V_{OUT} = 4V/1V$, $I_{load} = 100mA/phase$
Transient performance	$V_{DROP} = 136 \text{ mV}$, Recovery time = 880 ns @ $I_{load} = 0.2A$ to 4.2 A (4A/200 ns), $V_{IN}/V_{OUT} = 4V/1V$

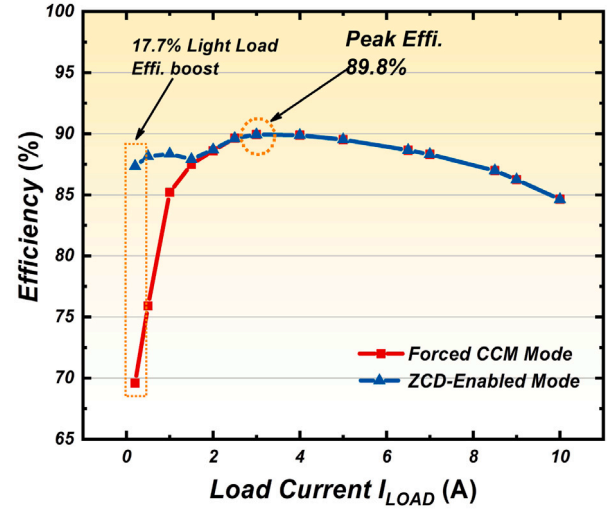


Fig. 14. Measured power conversion efficiency versus load current with and without the ZCD enabled.

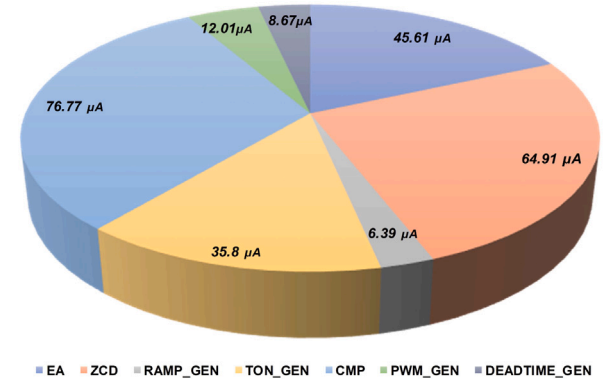


Fig. 15. Simulated quiescent current (I_q) breakdown of the proposed current-mode AOT controller operating in deep DCM.

transitions between DCM and CCM during drastic load changes. This real-time ZCD capability ensures natural frequency reduction during heavy-to-light load transitions, greatly boosting dynamic light-load efficiency without introducing significant voltage deviations or loop instability.

The measured power conversion efficiency across a wide range of load currents is plotted in Fig. 14. Operating at a nominal 6-MHz switching frequency with a substantial 4-to-1-V VCR, the prototype achieves a competitive peak efficiency of 89.8%. Moreover, the inclusion of the ZCD circuit, combined with the autonomous frequency reduction characteristic of the valley current-mode AOT controller, plays a critical role in low-power operation. Compared to a forced-CCM operation, the ZCD-enabled mode delivers a substantial 17.7% efficiency improvement under light-load conditions.

Table 3
Comparison with prior works.

Parameter	[18] JSSC '14	[4] TPEL '16	[6] TCAS-II '20	[7] TPEL '23	This work
Technology	180 nm BCD	28 nm	180 nm	65 nm	180 nm BCD
Topology	4-Phase Buck	1-Phase Buck	1-Phase Buck	3-Level Hybrid	2-Phase Buck
Control scheme	CM PWM/PFM	AOT	COT	AOT	Valley-Current AOT
VIN/VOUT (V)	2.7–5.0/0.8–VIN	3.3/1.05	3.3/1.03	2.7–4.5/0.6–1.3	2.5–5.0/0.3–1.2
FSW (MHz)	2	2.5	1	2.5	6
Filter (L/COUT)	470 nH/44 μ F	1 μ H/4.7 μ F	10 μ H/10 μ F	470 nH/10 μ F	120 nH/8.4 μ F
Max load (A)	4.5	1.7	1	1.6	10
Peak efficiency	91.60%	89.00%	93.40%	88.60%	89.80%
Load step (A)	N.R.	1.4	0.9	1	3
Recovery time	N.R.	8 μ s	18 μ s	3.0 μ s	0.88 μ s
Norm. Recovery time	N.R.	5.71 μ s/A	20.0 μ s/A	3.0 μ s/A	0.22 μ s/A
Light-load scheme	Phase-shedding	N.R.	DCM	DCM	DCM

To substantiate the light-load efficiency improvements demonstrated in Fig. 14, the simulated quiescent current (I_q) breakdown of the proposed controller operating in deep DCM is presented in Fig. 15.

Under DCM operation, the switching frequency scales down proportionally with the load current, resulting in extended standby periods between consecutive switching pulses. To minimize the standby power overhead, the proposed controller employs an adaptive bias-shutdown strategy. During these prolonged idle phases, the bias current sources for non-essential timing and logic modules are completely disabled. For example, the zero-crossing detector (ZCD) incorporates a self-power-down mechanism. Upon detecting the inductor current zero-crossing and truncating the reverse current, the ZCD immediately cuts off its own bias to save power until the next switching cycle.

By implementing this dynamic bias management, the total static current of the controller is strictly limited to 250.2 μ A (equivalent to <1 mW standby power). As illustrated in Fig. 15, the remaining power budget is primarily allocated to the critical modules: the main comparator (76.77 μ A), the ZCD (64.91 μ A), and the error amplifier (45.61 μ A). Sustaining sufficient bias current in the always-on CMP is physically imperative to ensure zero-latency load transient detection. This strategic trade-off—selectively shutting down idle internal circuit branches while preserving the wide bandwidth of critical detection modules—directly facilitates the excellent efficiency observed at the 200-mA light-load condition.

This significant efficiency boost thoroughly verifies that the proposed control architecture successfully mitigates high-frequency switching losses during standby modes, making it a highly viable and robust solution for power-constrained VRM application for smart devices.

The overall measured performance metrics of the fabricated prototype are comprehensively summarized in Table 2. Implemented in a standard 0.18- μ m BCD process, the fully integrated dual-phase controller and power stage occupy a silicon area of 2×2.4 mm². Under the nominal 4-V to 1-V VCR, the system secures a peak efficiency of 89.6% at a load of 1.5 A per phase. Bolstered by the auto-zeroing ZCD and autonomous DCM transitions, the converter maintains a highly competitive light-load efficiency of 88.3% at 500 mA per phase. For dynamic regulation, the architecture physically validates its extreme transient agility, restricting the maximum voltage droop (V_{DROP}) to a tight 136 mV and recovering within 880 ns under a severe 4-A step load (2 A/phase, 4 A/200 ns slew rate).

Table 3 benchmarks the proposed dual-phase AOT buck converter against recent state-of-the-art analog-controlled converters targeting direct Li-ion battery inputs [18–21]. Operating at a 6 MHz high switching frequency, the proposed design aggressively miniaturizes the passive filter, requiring only a miniature 120-nH inductor and an 8.4- μ F output capacitor to cleanly support a massive 10-A load. In contrast, conventional 1–2.5 MHz ACOT and COT designs rely on bulkier 0.47–10 μ H inductors to maintain stability for significantly lower load capacities [18,20,21]. Furthermore, the valley-current AOT controller fundamentally circumvents traditional loop-bandwidth limitations of

fixed frequency controller, achieving fast transient responses. Because comparing absolute recovery times is inherently biased by varying load-step magnitudes, a normalized recovery time metric (μ s/A) is evaluated to ensure a relatively rigorous and fair comparison. Under a severe 4-A load step, the proposed converter settles in just 0.88 μ s, translating to 0.22 μ s/A. This represents an order-of-magnitude improvement over recent architectures that exhibit significantly slower normalized dynamics ranging from 3.0 μ s/A to 20.0 μ s/A [18,20,21]. Finally, instead of employing logic-heavy phase-shedding schemes [19], this work integrates a high-precision ZCD comparator for autonomous DCM transitions, yielding a 17.7% efficiency boost at light loads and maintaining an 89.80% peak efficiency without computational overhead. Ultimately, the proposed converter successfully breaks the traditional trade-offs between switching frequency, component volume, and transient bandwidth, presenting a highly compact and robust power management solution for modern SoCs.

6. Conclusion

In this paper, a highly compact and robust 6-MHz dual-phase AOT buck converter is proposed and experimentally verified for Li-ion battery-powered SoC VRM applications. By employing a virtual inductor current sensing network, the controller bypasses the gain-bandwidth product (GBW) constraints of active amplifiers, acquiring clean and accurate valley current information even in severe PDN noise environments. A deterministic, PLL-less phase manager effectively distributes the modulated pulses with a rigorous 180° phase shift, securing optimal ripple cancellation while maintaining wideband small-signal stability. Moreover, the integration of a two-stage auto-zeroing ZCD provides an ultra-fast, sub-60-mA precise turn-off mechanism for the low-side switches, facilitating autonomous and energy-efficient transitions between CCM and DCM to significantly extend battery run-times. Experimental results demonstrate that the prototype achieves an outstanding normalized transient recovery time of 0.22 μ s/A under a 4-A load step, significantly outperforming conventional analog-controlled converters. Coupled with a peak efficiency of 89.8% and a 17.7% light-load efficiency boost, the proposed architecture successfully resolves the critical design bottlenecks of high-frequency multiphase converters. Ultimately, it presents a highly viable, extremely compact, and robust power management solution for next-generation Li-ion battery-driven smart devices.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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